

1. Which functional group is present in alcohols?

- a) Hydroxyl
- b) Carbonyl
- c) Carboxyl
- d) Amino

Answer: a

2. What is the pH of a neutral solution?

- a) 7
- b) 0
- c) 14
- d) 1

Answer: a

3. Which bond joins atoms in an organic molecule?

- a) Covalent bond
- b) Ionic bond
- c) Metallic bond
- d) Hydrogen bond

Answer: a

4. Which element is the backbone of organic chemistry?

- a) Carbon
- b) Oxygen
- c) Nitrogen
- d) Sulfur

Answer: a

5. Which of these is a strong acid?

- a) Hydrochloric acid
- b) Acetic acid
- c) Citric acid
- d) Carbonic acid

Answer: a

6. What is a mole?

- a) Amount of substance containing Avogadro's number of particles
- b) A unit of mass
- c) A unit of volume
- d) A type of bond

Answer: a

7. Which gas is produced when an acid reacts with a metal?

- a) Hydrogen
- b) Oxygen
- c) Nitrogen
- d) Carbon dioxide

Answer: a

8. Which isomerism arises from different spatial arrangements?

- a) Stereoisomerism
- b) Structural isomerism
- c) Tautomerism
- d) Metamerism

Answer: a

9. Which process involves a substance losing electrons?

- a) Oxidation
- b) Reduction
- c) Neutralization
- d) Precipitation

Answer: a

10. Which of these is an example of a buffer solution?

- a) Acetic acid and sodium acetate
- b) Pure water
- c) Sodium chloride solution
- d) Dilute hydrochloric acid

Answer: a

11. Explain the difference between an acid and a base with examples.

(Write your answer in the space provided.)

12. Describe the structure of the benzene ring and its significance in drug chemistry.

(Write your answer in the space provided.)